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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCS132 - BUSINESS INTELLIGENCE

(For MCS)

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Construct a diagram to show characteristics, sources and challenges associated with unstructured data.	8	2	1	1
b) Summarize the Unstructured Information Management Architecture (UIMA) and its applicability to handle unstructured data with diagram	6	2	1	1
c) Interpret all stages of ETL process with a diagram	6	2	2	1
2. a) Identify and compare the approaches for building data warehouses with diagram	10	3	3	1
b) Interpret the various program specific and project specific roles and their responsibilities in business intelligence	10	2	2	1
3. a) Identify and compare the key features of OLTP (Online Transaction Processing) and OLAP (Online Analytical Processing) systems	10	3	1	1
b) Outline the design of a Business Intelligence (BI) component framework diagram. Elaborate various functions of all layers within this framework.	10	2	2	1

Module - II

4. a) Construct snowflake multidimensional modeling with an example and identify its working and describe the same.	10	3	3	1,2
b) Demonstrate slice, dice, pivot, drill-up operations on a data cube with through an example.	10	2	3	1,2
5. a) What is data integration? Discuss its need and types with examples	10	2	4	1
b) Construct the architecture of a Federated Database and identify its components and describe in detail the same.	10	3	4	1
6. a) What are dashboards? Describe various types of dashboards with examples	10	2	4	1
b) Show four perspectives of a balanced scorecard. Identify four steps for creating balanced scorecard using a flowchart and describe the same.	10	3	4	1

Module - III

7. a) Discuss Apriori algorithm step by step for frequent item set retrieval with an example.	10	2	5	1,2
b) Interpret all data mining functionalities in detail.	10	2	5	1
8. a) Summarize classification, clustering, outlier detection for text mining applications.	10	2	5	1
b) Illustrate the concept of association rules with suitable transaction data.	10	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
